Arista Access Node KR-ACC-24 Hardware/Environmental Fault Resolution SOP (ALARM_7273)

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_7273

Description

This Standard Operating Procedure outlines the steps to diagnose and resolve hardware/environmental faults on the Arista Access Node KR-ACC-24, specifically related to ASIC memory parity errors, line card crashes, and power supply redundancy loss. The Arista 7250X series switches utilize a hardware-based, ASIC-driven architecture to process and forward packets. In the event of a hardware fault, the ASIC's on-die memory parity checking mechanism may fail, resulting in packet corruption or loss. Furthermore, line card failures can lead to packet forwarding disruptions, while power supply redundancy loss may cause the node to shut down. This SOP focuses on isolating and resolving these hardware-related issues to ensure optimal network performance and reliability. Through a combination of diagnostic commands and resolution steps, network administrators can quickly identify and address the root cause of the fault, minimizing downtime and ensuring continuous network operation.

Pre-requisite Checks

- Prerequisite: Verify the node is accessible and reachable via SSH or console

- Prerequisite: Check the redundant link status to ensure link redundancy is maintained

- Prerequisite: Confirm that all power supplies are online and functioning correctly

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource, Expected Output: CPU, memory, and ASIC resource utilization levels, Description: This command provides insight into the node's system resource utilization, helping identify potential issues with ASIC memory parity errors or line card crashes

Command: show interface status, Expected Output: Interface status, including link speed and duplex, Description: This command isolates potential issues related to line card crashes or power supply redundancy loss, as it provides a detailed view of the node's interface status
```

```
Command: show power supply, Expected Output: Power supply status, including online and redundant status, Description: This command verifies the power supply redundancy status, helping diagnose potential issues with power supply redundancy loss
```

Resolution Steps

Step 1: Step: Replace the failing ASIC or line card, Command: system reboot, Description: This step involves physically replacing the failing ASIC or line card with a functional one, followed by a system reboot to ensure the node is operational

Step 2: Step: Restore power supply redundancy, Command: power supply insert , Description: This step involves physically inserting a redundant power supply into the node and configuring it online to restore power supply redundancy

Post-Check Verification

```
Command: show system resource, Expected Output: CPU, memory, and ASIC resource utilization levels, Description: This command verifies that the node's system resource utilization has returned to normal levels

Command: show interface status, Expected Output: Interface status, including link speed and duplex, Description: This command verifies that the node's interface status has returned to normal

Command: show power supply, Expected Output: Power supply status, including online and redundant status, Description: This command verifies that the power supply redundancy status has been restored
```

Roll-back Procedures

- Step: Restore previous power supply configuration, Command: power supply delete , Description: This step involves deleting the redundant power supply and restoring the previous power supply configuration to ensure the node remains operational

- Step: Revert system configuration changes, Command: system revert, Description: This step involves reverting any system configuration changes made during the resolution process to ensure the node remains operational